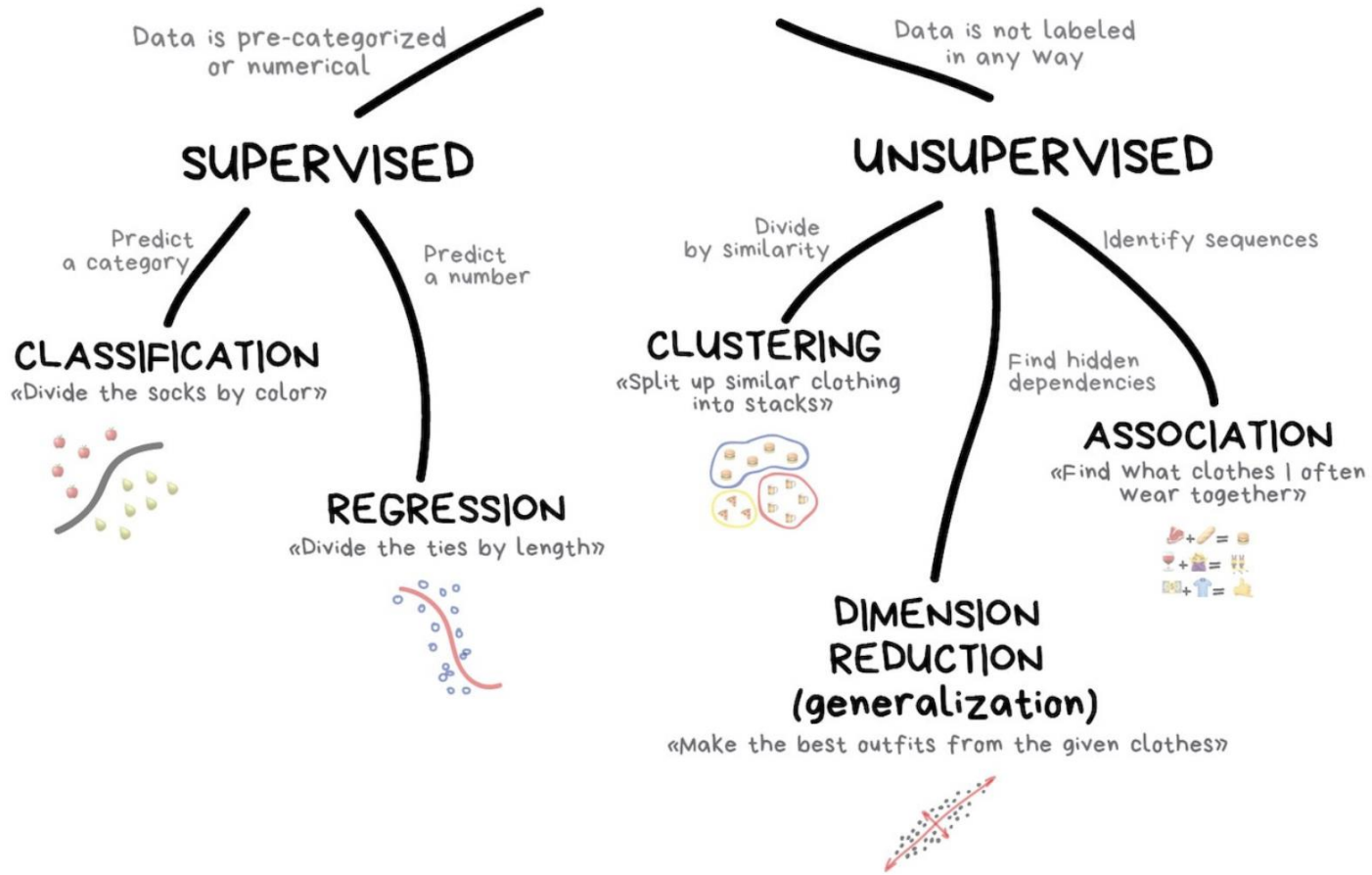
Association Rule Discovery: การสืบค้นกฎความสัมพันธ์ and its application to Data Analytics

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Classical Machine Learning



https://vas3k.com/blog/machine_learning/?ref=hn

Association Rule Discovery: การสืบค้นกฎความสัมพันธ์

- Example of Association Rules

  {Diaper} → {Beer}

- ″If a person buys Diapers, they tend to also buy Beer″
- **Left** = what already purchased
- **Right** = what tends to be purchased along with it
- Also known as **Market Basket Analysis**

Transactional Database

TID	*Items*
1	Bread, Milk
2	Bread, Diaper, Beer, Eggs
3	Milk, Diaper, Beer, Coke
4	Bread, Milk, Diaper, Beer
5	Bread, Milk, Diaper, Coke

Example of Association Rules

{Diaper} → {Beer},
{Milk, Bread} → {Eggs,Coke},
{Beer, Bread} → {Milk},

Association Rules

User ID	Movies liked
46578	Movie1, Movie2, Movie3, Movie4
98989	Movie1, Movie2
71527	Movie1, Movie2, Movie4
78981	Movie1, Movie2
89192	Movie2, Movie4
61557	Movie1, Movie3

Potential Rules:

Movie1 ⇨ Movie2

Movie2 ⇨ Movie4

Movie1 ⇨ Movie3

Association Rule Discovery: Market Basket Analysis

INPUT = **Transactions**,

OUTPUT = **Rules** that will predict the occurrence of an item based on the occurrences of other items in the transaction

Transactional Database

TID	*Items*
1	Bread, Milk
2	Bread, Diaper, Beer, Eggs
3	Milk, Diaper, Beer, Coke
4	Bread, Milk, Diaper, Beer
5	Bread, Milk, Diaper, Coke

Example of Association Rules

{Diaper} → {Beer},
{Milk, Bread} → {Eggs,Coke},
{Beer, Bread} → {Milk},

Association Rule Discovery: การสืบค้นกฎความสัมพันธ์



People who bought …
also bought …





Association Rule Discovery: Market Basket Analysis

Customer buying habits by finding **associations between the different items** that customers place in their “shopping basket”



Association Rules: Relationship between Items

Association Rule is the relationship between items that frequently appear together in transactions (e.g., purchases)

Example: "Customers who buy 'Diapers' tend to also buy 'Beer'"

This is a technique that falls under the category of **Data Mining** (Market Basket Analysis).

TID	*Items*
1	Bread, Milk
2	Bread, Diaper, Beer, Eggs
3	Milk, Diaper, Beer, Coke
4	Bread, Milk, Diaper, Beer
5	Bread, Milk, Diaper, Coke

Transactional Database

Example of Association Rules

$\{Diaper\} \rightarrow \{Beer\}$

Structure of rules

A rule in the form **X** → **Y** means:

"If a customer buys X, they tend to also buy Y"

X = Condition (what was already purchased)

Y = Frequent consequence (what should be recommended or promoted)

TID	*Items*
1	Bread, Milk
2	Bread, Diaper, Beer, Eggs
3	Milk, Diaper, Beer, Coke
4	Bread, Milk, Diaper, Beer
5	Bread, Milk, Diaper, Coke

Transactional Database

{Diaper} → {Beer},
{Milk, Bread} → {Eggs,Coke},
{Beer, Bread} → {Milk}

Benefits of Association Rules

- **Marketing / Retail:**
  - Automatic Product Recommendation
  - Planning Bundle Promotions
  - Arranging store product placement to encourage co-purchases
- **Digital:**
  - Creating paired product content on TikTok or Instagram
  - Improving Retargeting Ads accuracy
- **Others:**
  - Fraud Detection
  - Analyzing website user behavior

Association Rules Applications

- **Retail marketing**:
  - Targeted Promotion
  - Store layout and product placement
  - Personalized Recommendation
- **Fraud detection**: Identifying transactions that occur in unusual locations or at unusual times.
- **Medical diagnosis**: Identify relationships between symptoms and diseases to improve the accuracy of medical diagnoses.
- **Web usage mining**: Identify patterns in how people use websites to improve the design and usability of websites.
- **Network intrusion detection**: Identify patterns of network traffic that may be indicative of an attack.
- **Etc**.

Some Basic Definitions

- **Itemset**
  - A collection of one or more items
    - Example: {Milk, Bread, Diaper}
  - k-itemset
    - An itemset that contains k items
- **Support count (σ)**
  - Frequency of occurrence of an itemset
  - E.g. σ({Milk, Bread,Diaper}) = 2
- **Support**
  - Fraction of transactions that contain an itemset
  - E.g. s({Milk, Bread, Diaper}) = 2/5 = 40%
  - Transactions 4,5 contain {Milk, Bread, Diaper}
- **Frequent Itemset**
  - An itemset whose support is greater than or equal to a *minsup* threshold.
  - If *minsup* = 40%
  - Then {Milk, Bread, Diaper} is a Frequent Itemset.

TID	*Items*
1	Bread, Milk
2	Bread, Diaper, Beer, Eggs
3	Milk, Diaper, Beer, Coke
4	Bread, Milk, Diaper, Beer
5	Bread, Milk, Diaper, Coke

Association Rule: Evaluation Measures

- **Association Rule**
  - An implication expression of the form $X \rightarrow Y$, where X and Y are itemsets
  - Example:
    - {Diaper} → {Beer}
    - {Milk, Diaper} → {Beer, Coke}
    - Etc.

TID	Items
1	Bread, Milk
2	Bread, Diaper, Beer, Eggs
3	Milk, Diaper, Beer, Coke
4	Bread, Milk, Diaper, Beer
5	Bread, Milk, Diaper, Coke

- **Rule Evaluation Measures**
  - Confidence (X->Y)
    - What is the probability of buying Y if he already bought X?
    - Measures how often items in Y appear in those transactions that contain X
  - Support (X->Y)
    - Fraction of transactions that contain both X and Y

Example:

$$\{\text{Milk}, \text{Diaper}\} \Rightarrow \text{Beer}$$

$$c = \frac{\sigma(\text{Milk}, \text{Diaper}, \text{Beer})}{\sigma(\text{Milk}, \text{Diaper})} = \frac{2}{3} = 0.67$$

$$s = \frac{\sigma(\text{Milk}, \text{Diaper}, \text{Beer})}{|\text{T}|} = \frac{2}{5} = 0.4$$

Association Rule: Evaluation Measures

- **Rule X -> Y**
- **Support:**
  - What % of all transactions contain both X and Y
  - Example: 100 transactions → 30 contain {Diaper + Beer} → Support = 30%
- **Confidence:**
  - Of all customers who bought X, what % also bought Y
  - Example: 50 customers bought Diaper, 40 also bought Beer → Confidence = 80%
- **Lift:**
  - The true strength of the relationship (compared to random chance)
  - Lift > 1 means "there is a genuine relationship / true dependency"

Association Rule Discovery

Trans-ID	Items
1	ACD
2	BCE
3	ABCE
4	BE
5	ABCE

Min_support 60%
Min_confidence 75%

Rule	Conf.
{BC} =>{E}	100%
{BE} =>{C}	75%
{CE} =>{B}	100%
{B} =>{CE}	75%
{C} =>{BE}	75%
{E} =>{BC}	75%

Some Rules

Support(X => Y) = Support($X \cup Y$)
Confidence(X => Y) = Support($X \cup Y$)/ Support(X)

Association Rule Discovery

TID	*Items*
1	Bread, Milk
2	Bread, Diaper, Beer, Eggs
3	Milk, Diaper, Beer, Coke
4	Bread, Milk, Diaper, Beer
5	Bread, Milk, Diaper, Coke

Example of Rules:

{Milk,Diaper} → {Beer} (s=0.4, c=0.67)
{Milk,Beer} → {Diaper} (s=0.4, c=1.0)
{Diaper,Beer} → {Milk} (s=0.4, c=0.67)
{Beer} → {Milk,Diaper} (s=0.4, c=0.67)
{Diaper} → {Milk,Beer} (s=0.4, c=0.5)
{Milk} → {Diaper,Beer} (s=0.4, c=0.5)

Association Rule Discovery

TID	Items
1	Bread, Milk
2	Bread, Diaper, Beer, Eggs
3	Milk, Diaper, Beer, Coke
4	Bread, Milk, Diaper, Beer
5	Bread, Milk, Diaper, Coke

Example of Rules:

{Milk,Diaper} → {Beer} (s=0.4, c=0.67)
{Milk,Beer} → {Diaper} (s=0.4, c=1.0)
{Diaper,Beer} → {Milk} (s=0.4, c=0.67)
{Beer} → {Milk,Diaper} (s=0.4, c=0.67)
{Diaper} → {Milk,Beer} (s=0.4, c=0.5)
{Milk} → {Diaper,Beer} (s=0.4, c=0.5)

- All the above rules are generated from {Milk, Diaper, Beer}
- They all have the same support => 0.4

Ex1: Generate Rules from {A,B,E}

Given frequent set {A,B,E}, what are association rules have

- minsup = 2

- minconf= 50% ?

A, B => E : conf=2/4 = 50%

TID	List of items
1	**A, B, E**
2	B, D
3	B, C
4	**A, B**, D
5	A, C
6	B, C
7	A, C
8	**A, B**, C, E
9	**A, B**, C

Ex2: Generate Rules from {A,B,E}

Given frequent set {A,B,E}, what association rules have minsup = 2 and minconf= 50% ?

A, B => E : conf=2/4 = 50%

A, E => B : conf=2/2 = 100%

B, E => A : conf=2/2 = 100%

E => A, B : conf=2/2 = 100%

Don't qualify

A =>B, E : conf=2/6 =33%< 50%

B => A, E : conf=2/7 = 28% < 50%

__ => A,B,E : conf: 2/9 = 22% < 50%

TID	List of items
1	**A, B, E**
2	B, D
3	B, C
4	A, B, C, D
5	A, C
6	B, C
7	A, C
8	**A, B, C, E**
9	A, B, C

Generating Association Rules

Trans-ID	Items
1	ACD
2	BCE
3	ABCE
4	BE
5	ABCE

Min_support 60%
Min_confidence 75%

Association Rule Discovery

Apriori

Fp-growth

Eclat

Two-step approach:

1. Find all the Frequent Itemset
   - Generate all itemsets whose support ≥ minsup

2. Generate Rules using all the Frequent Itemsets
   - For each frequent itemset,
   - Generate high confidence rules, where each rule is a binary partitioning of a frequent itemset

Generating Association Rules from Frequent Itemsets

Trans-ID	Items
1	ACD
2	BCE
3	ABCE
4	BE
5	ABCE

Min_support 60%
Min_confidence 75%

Frequent Itemset	Support
{**BCE**},{AC}	60%
{BC},{CE},{A}	60%
{BE},{B},{C},{E}	80%

Rule	Conf.
{BC} =>{E}	100%
{BE} =>{C}	75%
{CE} =>{B}	100%
{B} =>{CE}	75%
{C} =>{BE}	75%
{E} =>{BC}	75%

Support(X => Y) = Support($X \cup Y$)
Confidence(X => Y) = Support($X \cup Y$)/ Support(X)

Association Rule Discovery

Python Apriori

Rapidminer-Fp-growth

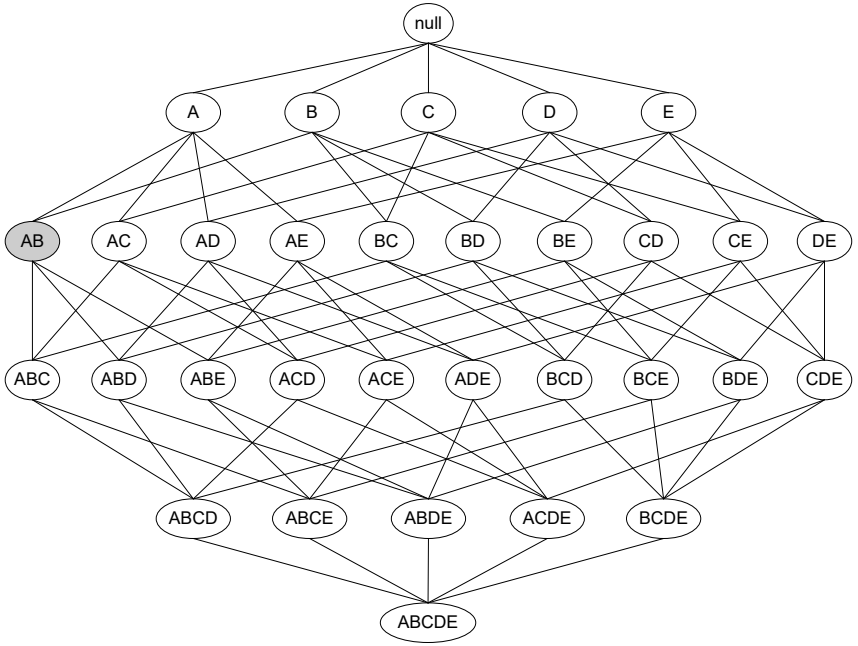
Eclat

- Two-step approach:
  1. Find all the Frequent Itemsets
     - Generate all itemsets whose support ≥ minsup
  2. Generate Rules using all the Frequent Itemsets
     - For each frequent itemset,
     - Generate high confidence rules, where each rule is a binary partitioning of a frequent itemset
- Frequent itemset generation is still computationally expensive

Generate all the frequent itemsets

Trans-ID	Items
1	ACD
2	BCE
3	ABCE
4	BE
5	ABCE

Min_support 60%
Min_confidence 75%



Correlation/Lift

- Association rules do not capture correlations between items
- Suppose
  - 90% customers buy coffee
  - 25% buy tea
  - 20% buy both
- The rule tea => coffee has high support and confidence
- But items tea and coffee are not correlated
  - Correlation/Lift = Conf/Expected Confidence
  - Correlation/Lift(X -> Y) = Conf(X -> Y)/Support(Y)
- Value < 1 indicates a negative correlation
- Value = 1 signifies independence
- Value > 1 indicates a positive correlation

Correlation/Lift Measure

TID	Items
1	A
2	A B
3	A B
4	A B
5	B

$$Lift: = \frac{Confidence}{ExpectedConfidence}$$

$$\text{lift}(M_1 \rightarrow M_2) = \frac{\text{confidence}(M_1 \rightarrow M_2)}{\text{support}(M_2)}$$

Example:

$$\{A\} \Rightarrow_{s,\alpha} \{B\}$$

$$s = \frac{\sigma(A,B)}{\text{Total Number of Transactions}} = \frac{3}{5} = 0.6$$

$$\alpha = \frac{\sigma(A,B)}{\sigma(A)} = \frac{3}{4} = 0.75$$

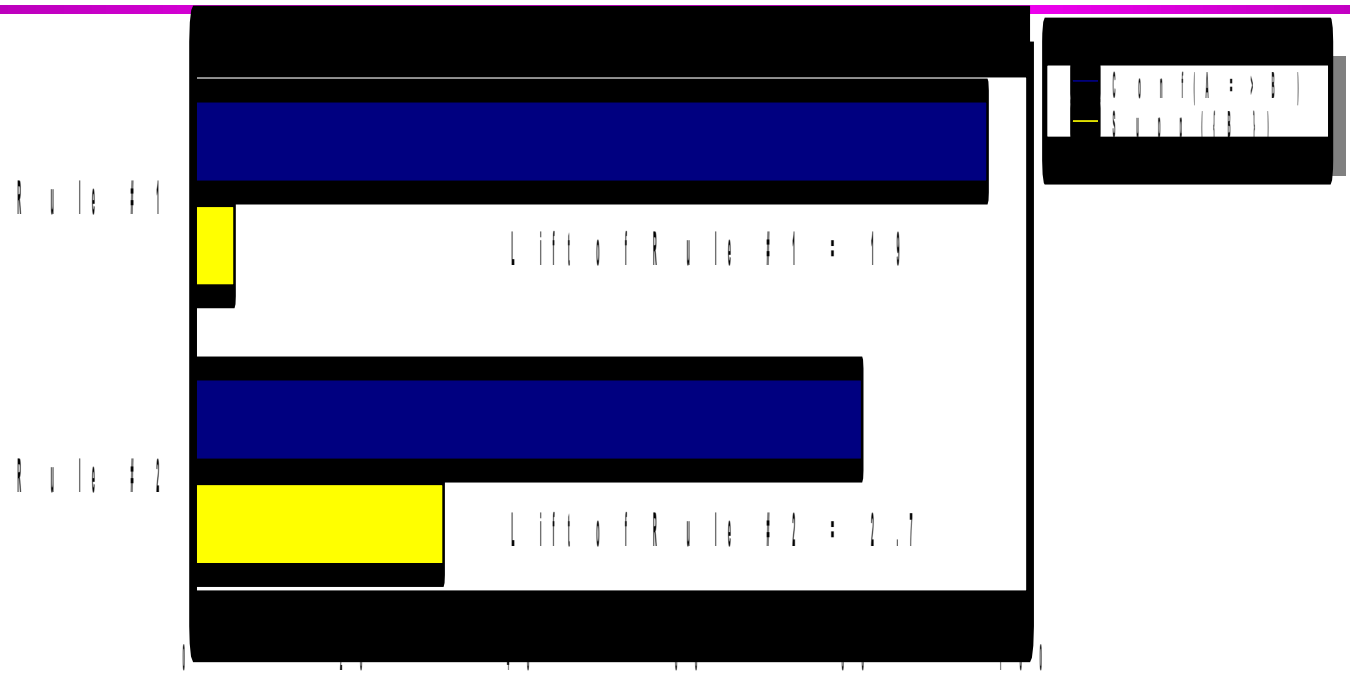
$$L = \frac{Confidence}{\text{Expected Confidence}} = \frac{0.75}{0.8} = 0.93$$

Lift = Confidence (A->B)/Support(B)

The probability of buying B (with the knowledge that A already bought) is much lower than the probability of buying B (without the knowledge that A already bought).

The fact that A already bought not increase the probability of buying B.

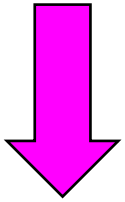
Lift of a Rule



Data-pre-processing for Transactional Databases

TID	*Items*
1	Bread, Milk
2	Bread, Diaper, Beer, Eggs
3	Milk, Diaper, Beer, Coke
4	Bread, Milk, Diaper, Beer
5	Bread, Milk, Diaper, Coke

Most of existing libraries for Association Rules work with Binary Format !



TID	Bread	Milk	Diaper	Beer	Eggs	Coke
1	1	1	0	0	0	0
2	1	0	1	1	1	0
3	0	1	1	1	0	1
4	1	1	1	1	0	0
5	1	1	1	0	0	1

Multi-dimensional AR

Associations between values of different attributes :

CID	**nationality**	**age**	**income**
1	Italian	50	low
2	French	40	high
3	French	30	high
4	Italian	50	medium
5	Italian	45	high
6	French	35	high

RULES:

nationality = French $\Rightarrow$ income = high [50%, 100%]

income = high $\Rightarrow$ nationality = French [50%, 75%]

age = 50 $\Rightarrow$ nationality = Italian [33%, 100%]

Quantitative Attributes

- Quantitative attributes (e.g. age, income)
- Categorical attributes (e.g. color of car)

CID	height	weight	income
1	168	75,4	30,5
2	175	80,0	20,3
3	174	70,3	25,8
4	170	65,2	27,0

Problem: too many distinct values

Solution: transform quantitative attributes in categorical ones via discretization.

Quantitative Association Rules

CID	Age	Married	NumCars
1	23	No	1
2	25	Yes	1
3	29	No	0
4	34	Yes	2
5	38	Yes	2

[Age: 30..39] and [Married: Yes] ⇒ [NumCars:2]

Min-support = 40%
Min-confidence = 50%

Discretization of quantitative attribuite

Solution: each value is replaced by the interval to which it belongs.

height: 0-150cm, 151-170cm, 171-180cm, >180cm

weight: 0-40kg, 41-60kg, 60-80kg, >80kg

income: 0-10ML, 11-20ML, 20-25ML, 25-30ML, >30ML

CID	height	weight	income
1	151-171	60-80	>30
2	171-180	60-80	20-25
3	171-180	60-80	25-30
4	151-170	60-80	25-30

Problem: the discretization may be useless (see weight).

Categorical Attributes: Example

Weather Condition	Driver's Condition	Traffic Violation	Seat Belt	Crash Severity
Good	Alcohol-impaired	Exceed speed limit	No	Major
Bad	Sober	None	Yes	Minor
Good	Sober	Disobey stop sign	Yes	Minor
Good	Sober	Exceed speed limit	Yes	Major
Bad	Sober	Disobey traffic signal	No	Major
Good	Alcohol-impaired	Disobey stop sign	Yes	Minor
Bad	Alcohol-impaired	None	Yes	Major
Good	Sober	Disobey traffic signal	Yes	Major
Good	Alcohol-impaired	None	No	Major
Bad	Sober	Disobey traffic signal	No	Major
Good	Alcohol-impaired	Exceed speed limit	Yes	Major
Bad	Sober	Disobey stop sign	Yes	Minor

- **Transform categorical attribute**

  => into binary variables (one-hot encoding)

- **Introduce a new "item" for each distinct attribute-value pair**

Handling Categorical Attributes => Transform to Binary

Weather Condition	Driver's Condition	Traffic Violation	Seat Belt	Crash Severity
Good	Alcohol-impaired	Exceed speed limit	No	Major
Bad	Sober	None	Yes	Minor
Good	Sober	Disobey stop sign	Yes	Minor
Good	Sober	Exceed speed limit	Yes	Major
Bad	Sober	Disobey traffic signal	No	Major
Good	Alcohol-impaired	Disobey stop sign	Yes	Minor
Bad	Alcohol-impaired	None	Yes	Major
Good	Sober	Disobey traffic signal	Yes	Major
Good	Alcohol-impaired	None	No	Major
Bad	Sober	Disobey traffic signal	No	Major
Good	Alcohol-impaired	Exceed speed limit	Yes	Major
Bad	Sober	Disobey stop sign	Yes	Minor

Good	Bad	Alcohol	Sober	Exceed speed	None	Disobey stop	Disobey traffic	Belt = No	Belt = Yes	Major	Minor
1	0	1	0	1	0	0	0	1	0	1	0
0	1	0	1	0	1	0	0	0	1	0	1
1	0	0	1	0	0	1	0	0	1	0	1
1	0	0	1	1	0	0	0	0	1	1	0
0	1	0	1	0	0	0	1	1	0	1	0
1	0	1	0	0	0	1	0	0	1	0	1
0	1	1	0	0	1	0	0	0	1	1	0
1	0	0	1	0	0	0	1	0	1	1	0
1	0	1	0	0	1	0	0	1	0	1	0
0	1	0	1	0	0	0	1	1	0	1	0
1	0	1	0	1	0	0	0	0	1	1	0
0	1	0	1	0	0	1	0	0	1	0	1

Case Study: BeautyCo

- **BeautyCo** wants to use purchase data to analyze:
- Which products customers frequently buy together ⟶ to plan **Predictive Analytics** (forecasting buying behavior)
- If they want to boost sales of certain products ⟶ which products should be paired in recommendation campaigns **(Prescriptive Analytics)**
- Use AI tools to discover association rules between products.

Media-Focused Column	**Media Context**
Travel_Pouch	สินค้าสำหรับ Travel Beauty Content
Cheek_Color	สินค้าสำหรับ Everyday Makeup Tutorials
Nail_Art	สินค้าสำหรับ Nail Art Influencer Campaigns
Makeup_Tools	สินค้าสำหรับ Professional Makeup Content
Coverage_Base	สินค้าสำหรับ Problem-Solving Beauty Content
Brow_Definer	สินค้าสำหรับ Brow Transformation Content
Contouring	สินค้าสำหรับ Contouring Tutorials
Lip_Liner	สินค้าสำหรับ Lip Art Content
Eye_Enhancer	สินค้าสำหรับ Eye Makeup Tutorials
Eye_Palette	สินค้าสำหรับ Creative Eye Looks
Base_Makeup	สินค้าสำหรับ Foundation Matching Content
Glossy_Lips	สินค้าสำหรับ Glossy Lip Trends
Lip_Color	สินค้าสำหรับ Lip Color Campaigns
Eye_Liner	สินค้าสำหรับ Eyeliner Art Content

EDA

There are a total of **1,000 transactions** and **14 product types**.
Each row = 1 customer's purchase transaction.

กลุ่มสินค้าเบส/รองพื้น (Base Products)

Base_Makeup - เบสเมคอัพ

Coverage_Base - รองพื้น

กลุ่มสินค้าริมฝีปาก (Lip Products)

Glossy_Lips - ลิปกลอส

Lip_Color - ลิปสติก

Lip_Liner - ลิปไลเนอร์

กลุ่มสินค้าตา (Eye Products)

Eye_Liner - อายไลเนอร์

Eye_Palette - พาเลทอายแชโดว์

Eye_Enhancer - อายเอนแฮนเซอร์

Brow_Definer - ดินสอเขียนคิ้ว

กลุ่มสินค้าแต่งแก้ม (Cheek Products)

Cheek_Color - บลัชออน

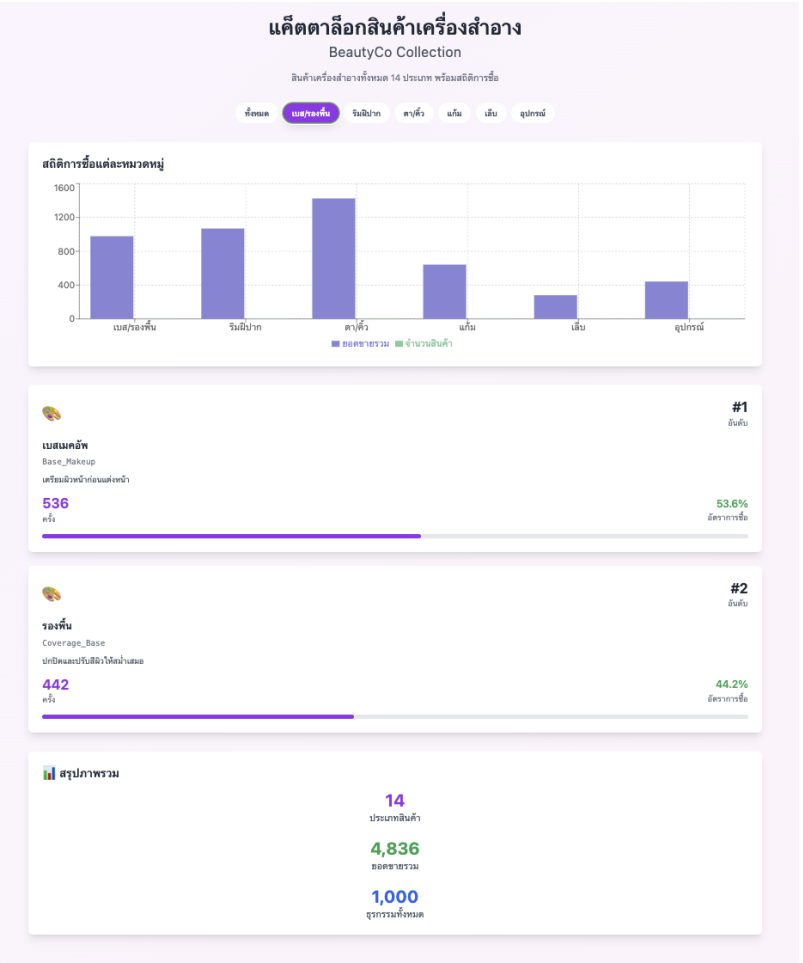
Contouring – คอนทัวร์

กลุ่มสินค้าเล็บและอุปกรณ์ (Nails & Tools)

Nail_Art - เนลอาร์ต

Makeup_Tools - เครื่องมือแต่งหน้า

Travel_Pouch – กระเป๋าเครื่องสำอาง



- **Observations:**
- **Base/Foundation** group has the highest sales rate (average 48.90%)
- **Eye & Brow** group has the most product variety (4 items)
- **Tools/Accessories** group has the lowest sales rate (average 21.73%)
- **Basic makeup** products are more popular than supplementary/add-on products

Item Frequency analysis

- From the **Item Frequency Analysis (Frequent Itemsets)**, the top 3 most purchased products are:

  #1: Base_Makeup

  - Purchases: 536 out of 1,000 transactions
  - Percentage: 53.60%

  #2: Glossy_Lips

  - Purchases: 490 out of 1,000 transactions
  - Percentage: 49.00%

  #3: Eye_Liner

  - Purchases: 457 out of 1,000 transactions
  - Percentage: 45.70%

- These are **very common cosmetic items**, especially **eye** and **base makeup**.
- So we would **expect combinations among eye products**.

Item Frequency analysis

Strong but expected combinations

- Some pairs have high support but are **not surprising** because they belong to the same makeup routine.

Pair	Support	Interpretation
Eye_Liner + Coverage_Base	0.297	Normal daily makeup routine
Eye_Palette + Eye_Enhancer	0.321	Eye makeup products used together
Coverage_Base + Eye_Palette	0.201	Typical full-face makeup

Potentially unexpected combinations

- These combinations stand out because they mix **different cosmetic categories**.
- **Base_Makeup + Glossy_Lips (0.356)**
- But this pair appears **quite frequently**

=> Suggests a **specific makeup style** where people apply **base + glossy lips together**

- This may indicate:
  - a **trend**
  - a **bundle opportunity**

Association Rules Generation

- **Parameter Settings:**
  - Using **FP-Growth** with min support = 0.2
  - Using **Create Association Rules** with min confidence = 0.7
- **Example Results:**
  - Number of frequent itemsets found: **23 sets**
- Maximal frequent itemsets:

No. of Sets: 12
Total Max. Size: 2

Min. Size: 2
Max. Size: 2
Contains Item:

Update View

Size	Support	Item 1	Item 2
2	0.356	Base_Makeup	Glossy_Lips
2	0.238	Base_Makeup	Eye_Liner
2	0.231	Base_Makeup	Coverage_Base
2	0.211	Base_Makeup	Eye_Palette
2	0.227	Glossy_Lips	Eye_Liner
2	0.218	Glossy_Lips	Coverage_Base
2	0.201	Glossy_Lips	Eye_Palette
2	0.297	Eye_Liner	Coverage_Base
2	0.201	Coverage_Base	Eye_Palette
2	0.220	Coverage_Base	Cheek_Color
2	0.204	Coverage_Base	Eye_Enhancer
2	0.321	Eye_Palette	Eye_Enhancer

- Example Rules:

No.	Premises	Conclusion	Support	Confidence	Lift ↓
2	Eye_Palette	Eye_Enhancer	0.321	0.843	2.360
3	Eye_Enhancer	Eye_Palette	0.321	0.899	2.360
1	Glossy_Lips	Base_Makeup	0.356	0.727	1.355

- It means that in any transaction containing **Eye_Palette**, there is an **84.3% chance** that **Eye_Enhancer** will also appear in that same transaction. This co-occurrence happens in **32.1%** of all transactions.

Interpretation and Application

- Rule with **highest Confidence**:

No.	Premises	Conclusion	Support	Confide... ↓	Lift
3	Eye_Enhancer	Eye_Palette	0.321	0.899	2.360

- **Meaning:** Customers who buy "Eye Enhancer" have an **89% likelihood** of also buying "Eye Shadow Palette" — a relationship stronger than random chance **(Lift > 1)**
- **Marketing Approach:**
  - Create a **"2 Styles Glam Eye Look"** promotion bundling **Eye Enhancer** and **Eye Shadow Palette**
  - Arrange **Influencer reviews** on **TikTok** or **Instagram Reels**

Interpretation and Application

Q&A

- **What does Confidence = 80% mean?**
  - It means that for every 100 transactions containing product group X, 80 of those transactions also include product group Y.
- **What does Support = 70% mean?**
  - It means that 70% of all transactions contain the items related to that rule.
- **Why is Lift > 1 useful?**
  - Because it indicates that buying X together with Y is not due to random chance — it reflects a relationship that is stronger than average.
- **If you were a Media Planner, how would you use these rules?**
  - Use rules with **high confidence & lift** to select products for paired content — for example, recommending a Lip + Gloss set through Influencers and running joint promotions.

Marketing Recommendations

- Create promotions for the **Top 3** best-selling products
- Bundle a **Basic Makeup Set** (Base + Lips + Eyes)
- Produce **TikTok content** showcasing usage techniques
- Develop and reposition **low-selling products** to make them more appealing